

Bootstrap 5

- ⊛ * Bootstrap is a free front-end framework for faster and easier web development.
 - * Bootstrap includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many other, as well as optional Javascript plugins.
 - * Bootstrap also gives you the ability to easily create responsive designs.
- ⊛ Why use Bootstrap;
- * Easy to use.
 - * Responsive features.
 - * Mobile-first approach.
 - * Browser compatibility.

⊛ Containers,

- * bootstrap 5 requires a containing element to wrap site contents.
- 1) The `.container` class provides a responsive 'fixed width container'.
- 2) The `.container-fluid` class provides a 'full width container', spanning the entire width of the viewport.

Fixed Container

- ⊛ Used the `.container` class to create a responsive, fixed-width container.

Fluid Container

- ⊛ Use the `.container-fluid` class to create a full width container, that will always span the entire width of the screen. (width is always 100%)

E.g. 1) `<div class="container">`

```
  <h1> My First Bootstrap Page </h1>  
  <p> This is some text. </p>
```

```
</div>
```

2) `<div class="container-fluid">`

```
  <h1> My first Bootstrap Page </h1>  
  <p> This is some text. </p>
```

```
</div>
```

Container Padding

- ★ By default, containers have left and right padding, no top or bottom padding. So; we can use "spacing utilities", such as extra padding and margins to make them look even better.
- ★ `.pt-5` means "add a large top padding".
E.g. `<div class="container pt-5"></div>`

Container Border and Color

- ★ borders and colors are also used together with containers.
E.g. `<div class="container p-5 my-5 border"></div>`
`<div class="container p-5 my-5 bg-dark text-white"></div>`
`<div class="container p-5 my-5 bg-primary text-white"></div>`

Responsive Containers

- ★ You can also use the `.container-sm | md | lg | xl | xxl` classes to determine when the container should be responsive.
(e.g. `.container-sm`, `.container-md`, `.container-lg`, `.container-xl`, `.container-xxl`)

Grids

- ★ Bootstrap's grid system is built with flexbox and allows up to 12 columns across the page.
- ★ If you don't want to use all 12 columns individually, you can group the columns together to create wider columns.
- ★ The grid system is responsive, and the columns will re-arrange automatically depending on the screen size.
- ★ This grid system has six classes;
 - `col` - (extra small devices - screen width less than 576px)
 - `col-sm` - (small devices - screen width = or > 576px)
 - `col-md` - (medium devices - screen width = or > 768px)
 - `col-lg` - (large devices - screen width = or > 992px)
 - `col-xl` (xlarge devices - screen width = or > 1200px)
 - `col-xxl` (xxlarge devices - screen width = or > 1400px)

④ The classes above can be combined to create more dynamic and flexible layouts.

④ For create a row;

```
<div class="row">
```

Three equal columns

④ The following example shows how to create three equal-width columns, on all devices and screen widths;

E.g.

```
<div class="row">
```

```
<div class="col"> here .col </div>
```

```
<div class="col"> here .col </div>
```

```
<div class="col"> here .col </div>
```

```
</div>
```

Responsive columns

E.g.

```
<div class="row">
```

```
<div class="col-sm-3"> .col-sm-3 </div>
```

```
<div class="col-sm-3"> .col-sm-3 </div>
```

```
<div class="col-sm-3"> .col-sm-3 </div>
```

```
<div class="col-sm-3"> .col-sm-3 </div>
```

```
</div>
```

Two unequal responsive columns

E.g.

```
<div class="row">
```

```
<div class="col-md-4"> .col-md-4 </div>
```

```
<div class="col-md-8"> .col-md-8 </div>
```

```
</div>
```

Text / Typography

<h1> - <h6>

④ Bootstrap styles HTML headings with a bolder font-weight and a responsive font-size.

E.g.

```
<p class="h1"> h1 Bootstrap heading </p>
```

```
<p class="h2"> h2 Bootstrap heading </p>
```

```
<p class="h3"> h3 Bootstrap heading </p>
```

```
<p class="h4"> h4 Bootstrap heading </p>
```

```
<p class="h5"> h5 Bootstrap heading </p>
```

```
<p class="h6"> h6 Bootstrap heading </p>
```


Display headings

- ⊛ Display headings are used to stand out more than normal headings (larger font-size and lighter font-weight). There are six classes to choose from: .display-1 to .display-6

E.g. `<h1 class="display-1">Display 1 </h1>`

`<h1 class="display-3">Display 3 </h1>`

`<h1 class="display-6">Display 6 </h1>`

<small>

- ⊛ .small class is used to create a smaller, secondary text in any heading. `<small>`

E.g. ~~`<h1 class="small">`~~

`<h1> h1 heading <small> Secondary text </small> </h1>`

<mark>

- ⊛ `<mark>` ^{and} ~~code~~ .mark with a yellow background color and some padding.

E.g. `<p> Use the mark element <mark> highlight </mark> </p>`

<blockquote>

- ⊛ Add the .blockquote class to a `<blockquote>` when quoting blocks of content from another source. And when naming a source, like "from WWF's website", use the .blockquote-footer class.

E.g. `<blockquote class="blockquote">`

`<p> This is blockquoted text. </p>`

`<footer class="blockquote-footer"> From WWF's web. </footer>
</blockquote>`

<code>

- ⊛ Bootstrap styles the HTML `<code>` element.

E.g. `<p> The following HTML elements :`

`<code> span </code> ,`

`<code> section </code> defines a section. </p>`

<kbd>

① Style the HTML <kbd> element in following way;

E.g. <p> Use <kbd> ctrl + p </kbd> to open the Print dialog box. </p>

② More typography classes;

1) .lead class makes a paragraph stand out.

E.g. <p class="lead"> This paragraph "stand out:" </p>

2) .text-start class indicates left aligned text.

• text-end class indicates right aligned text.

• text-center class indicates center aligned text.

• text-nowrap class indicates no wrap text.

E.g. <p class="text-start"> Left-aligned text </p>

<p class="text-end"> Right-aligned text </p>

<p class="text-center"> Center-aligned text </p>

<p class="text-nowrap"> No wrap text </p>

3) .text-break class prevents long text from breaking layout.

E.g. <p class="text-break">

This text using text-break. This text using text-break. This text using text-break. This text using text-break. This text using text-break.

</p>

4) .text-decoration-none class removes the underline from a link.

E.g. Click here

5) .text-lowercase class indicates lowercased text.

• text-uppercase class indicates uppercased text.

• text-capitalize class indicates capitalized text.

E.g. <p class="text-lowercase"> Lowercased text </p>

<p class="text-uppercase"> Uppercased text </p>

<p class="text-capitalize"> Capitalized text </p>

6) .list-unstyled class removes the default list-style and left margin on list terms. (use with and)

7) .list-inline class place all list items on a single line.

(This use together with .list-inline-item on each elements)

E.g. `<ul class="list-inline">`

`<li class="list-inline-item"> Coffee </li>`

`<li class="list-inline-item"> Tea </li>`

`</ul>`

Text colors - Background colors

- ★ In bootstrap we can use classes for text colors, to provide "meaning through colors".
- ★ The classes are; `.text-muted`, `.text-primary`, `.text-success`, `.text-info`, `.text-warning`, `.text-danger`, `.text-secondary`, `.text-white`, `.text-dark`, `.text-body`, `.text-light`
- ★ We can also add 50% opacity for black or white with `.text-black-50` or `.text-white-50` classes.
- ★ we can add classes for background colors, such as; `.bg-primary`, `.bg-success`, `.bg-info`, `.bg-warning`, `.bg-danger`, `.bg-secondary`, `.bg-dark`, `.bg-light`
- ★ we can use `.text-bg-color` classes and Bootstrap will automatically handle the appropriate text color for each background color.

E.g. `<p class="text-bg-primary"> Hello </p>`

`<p class="text-bg-danger"> Hello </p>`

Tables

- ★ `.table` class adds "light padding and horizontal dividers".

E.g. `<table class="table">`

`<thead>`

`<tr>`

`<th> Firstname </th>`

`<th> Lastname </th>`

`</tr>`

`</thead>`

`<tbody>`

`<tr>`

`<td> John </td>`

`<td> Doe </td>`

`</tr>`

`</tbody>`

`</table>`

★ `.table-striped` class adds zebra-stripes to a table;

E.g. `<table class="table table-striped">`
`</table>`

★ `.table-bordered` class adds border on all sides of the table and cells.

★ `.table-hover` class adds a hover effect (grey background color) on table rows.

★ `.table-dark` class adds a black color background to the table.

★ Combine `.table-dark` and `.table-striped` to create dark, striped table.

★ Also can combine `.table-dark` and `.table-hover`.

★ `.table-borderless` class removes borders from the table.

★ We can use `.table-primary`, `.table-success`, `.table-danger`, `.table-info`,
`.table-warning`, `.table-active`, `.table-secondary`, `.table-light`, `.table-dark`
classes for colorful table rows.

★ `.table-sm` class makes the table smaller by cutting cell padding in half.

★ `.table-responsive` class adds a scrollbar to the table when needed.

Images

★ The `.rounded` class adds rounded corners to an image.

★ The `.rounded-circle` class adds ~~is~~ a circle shape to an image.

★ The `.img-thumbnail` class shapes the image to a thumbnail (bordered).

★ Float an image to the left with the `.float-start` class or to the right with `.float-end` class.

★ We can center an image by adding the utility classes `.mx-auto` (margin $\rightarrow$ auto) and `.d-block` (display $\rightarrow$ block) to the image.

E.g. ``

★ Create responsive images by adding an `.img-fluid` class to the `<img>` tag.
The image will then scale nicely to the parent element.

This `.img-fluid` class applies `max-width: 100%;` and `height: auto;`

Alerts

- ⊛ Bootstrap 5 provides easy ways to create predefined alert messages.
- ⊛ Alerts created with `.alert` class, followed by one of the contextual classes like `.alert-success`, `.alert-info`, `.alert-warning`, `.alert-danger`, `.alert-primary`, `.alert-secondary`, `.alert-light`, `.alert-dark`.

E.g.

```
<div class="alert alert-success">
  <strong>Success! </strong> Here the alert!
</div>
```

- ⊛ We can add the `.alert-link` class to any links inside the alert box to create "matching colored links".
- ⊛ To close the alert message, add a `.alert-dismissible` class to the alert container. Then add `class="btn-close"` and `data-bs-dismiss="alert"` to a link or a button element.

E.g.

```
<div class="alert alert-success alert-dismissible">
  <button type="button" class="btn-close" data-bs-dismiss="alert">
  </button>
  <strong>Success! </strong> Here the alert!
</div>
```

- ⊛ The `.fade` and `.show` classes adds a fading effect when closing the alert message.

E.g.

```
<div class="alert alert-success alert-dismissible fade show">
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
  <strong>Success! </strong> Here the alert!
</div>
```

Buttons

- ⊛ In bootstrap 5, we can use below classes combine with `.btn` class.
`.btn-primary`, `.btn-secondary`, `.btn-success`, `.btn-info`, `.btn-warning`,
`.btn-danger`, `.btn-dark`, `.btn-light`, `.btn-link`

E.g.

```
<button type="button" class="btn btn-dark"> Click Me! </button>
```

★ The button classes can be used on `<a>`, `<button>` or `<input>`.

- ⊛ We have eight outline / bordered buttons. Move the mouse over them to see an additional "hover" effect.

In this case we can use below classes combine with `.btn` class.

- btn-outline-primary , • btn-outline-success , • btn-outline-secondary ,
- btn-outline-info , • btn-outline-warning , • btn-outline-danger , • btn-outline-dark
- btn-outline-light

④ We can use • btn-lg class for large buttons or • btn-sm class for small buttons.

⑤ To create a block level button that spans the entire width of the parent element, use the • d-grid "helper" class on the parent element.

E.g. `<div class="d-grid">`

`<button type="button" class="btn btn-primary btn-block">Hello`

`</button>`

`</div>`

⑥ A button can be set to an active (appear pressed) or a disabled (unclickable) state.

* The • active makes a button appear pressed, the • disabled class makes a button visually appear disabled.

* And also disabled attribute makes a button unclickable.

E.g. `<button type="button" class="btn btn-primary active">Active </button>`

`<button type="button" class="btn btn-primary" disabled>Disable </button>`

⑦ We can add "spinners" to a button.

E.g. `<button class="btn btn-primary">`

`<span class="spinner-border spinner-border-sm">Loading... </span>`

`</button>`

`<button class="btn btn-primary">`

`<span class="spinner-border spinner-grow-sm">Loading... </span>`

`</button>`

Button groups

⑧ We can group a series of buttons together (on a single line) with the • btn-group class.

* Use a `<div>` element with class • btn-group to create a button group.

⑨ We can use • btn-group-vertical class to create a vertical button group. It is also use with `<div>` tags.

- ★ Add the `.progress-bar-animated` class to animate the progress bar.
E.g. `<div class="progress-bar progress-bar-striped progress-bar-animated">`
`</div>`

- ★ We can add multiple progress bars in the same progress bar. Try this;

E.g. `<div class="progress">`
`<div class="progress-bar bg-success" style="width:40%;">`
Free space
`</div>`
`<div class="progress-bar bg-success" style="width:10%;">`
Warning
`</div>`
`</div>`

Spinners

- ★ To create a spinner/loader, we can use `.spinner-border` class.

E.g. `<div class="spinner-border"></div>`

- ★ We can use any text color utilities to add a color to the spinner.

E.g. `<div class="spinner-border text-primary"></div>`

- ★ We can use the `.spinner-grow` class, if you want to spinner/loader 'grow' instead of 'spin'.

E.g. `<div class="spinner-grow text-muted"></div>`

- ★ Use `.spinner-border-sm` or `.spinner-grow-sm` to create a smaller spinner.

Pagination

- ★ If we have a web site with lot of pages, we may wish to add some sort of pagination to each page.

- ★ To create a basic pagination, we can add `.pagination` class to an `<ul>` element. Then add the `.page-item` to each `<li>` element and a `.page-link` class to each link inside `<li>`

E.g. `<ul class="pagination">`
`<li class="page-item"><a class="page-link" href="#">Click</a></li>`
`<li class="page-item"><a class="page-link" href="#">Click</a></li>`
`</ul>`

- ★ The `.active` class is used to "highlight" the current page.

E.g. `<ul class="pagination">`

`<li class="page-item"><a class="page-link" href="#">Click </a></li>`

`<li class="page-item"><a class="page-link" href="#">Click </a></li>`

`<li class="page-item active"><a class="page-link" href="#">Click </a></li>`
`</ul>`

- ⊛ The `.disabled` class is used for un-clickable links.

E.g. `<ul class="pagination">`

`<li class="page-item disabled"><a class="page-link" href="#">Click </a></li>`
`</ul>`

- ⊛ Pagination blocks can be sized to a larger or smaller size.

⊛ We can add `.pagination-lg` class for larger blocks or `.pagination-sm` class for smaller blocks.

E.g. `<ul class="pagination pagination-lg">`
...
`</ul>`

- ⊛ We can use `.justify-content-center` and `.justify-content-end` classes to align the pagination block.

E.g. `<ul class="pagination justify-content-center">`
...
`</ul>`

- ⊛ Another form for pagination is "breadcrumbs".

⊛ The `.breadcrumb` and `.breadcrumb-item` classes indicates the current pages location within a navigational hierarchy.

E.g. `<ul class="breadcrumb">`

`<li class="breadcrumb-item"><a href="#">Photos </a></li>`

`<li class="breadcrumb-item"><a href="#">Italy </a></li>`

`<li class="breadcrumb-item active"><a href="#">Rome </li>`

`</ul>`

List groups

- ⊛ Basic list group is an unordered list with list items.

To create this, use an `<ul>` element with class `.list-group` and `<li>` elements with class `.list-group-item`

E.g. `<ul class="list-group">`

`<li class="list-group-item">First item </li>`

`</ul>`

⊛ We can use .active class within to highlight the current item.

E.g. <ul class="list-group">

<li class="list-group-item active"> Active item

⊛ To create a list group with linked items, use <div> instead of and <a> instead of . Optionally, add the .list-group-item-action class if you want a grey background on hover.

E.g. <div class="list-group">

 Hello
</div>

⊛ The .disabled class adds a lighter text color to the disabled item. And when used on links, it will remove the hover effect.

E.g. <div class="list-group">

 Disabled item
</div>

⊛ Use the .list-group-flush class to remove some borders and rounded corners.
⊛ This class used inside an element.

E.g. <ul class="list-group list-group-flush">

...

⊛ Use the .list-group-numbered class to create list items with numbers in front of them.

E.g. <ol class="list-group list-group-numbered">

...

⊛ If you want to display list items horizontally instead of vertically, add the .list-group-horizontal class to .list-group

E.g. <ul class="list-group list-group-horizontal">

...

⊛ We can use these classes for coloring list items;

.list-group-item-success, .list-group-item-secondary, .list-group-item-info,
.list-group-item-warning, .list-group-item-danger, .list-group-item-primary,
.list-group-item-dark, .list-group-item-light

Cards

⊛ A card is a bordered box with some padding around its content. It includes options for headers, footers, content, colors, ... etc.

⊛ A basic card is created with `.card` class, and content inside the card has a `.card-body` class.

E.g.

```
<div class="card">
  <div class="card-body"> Basic card </div>
</div>
```

⊛ The `.card-header` class adds a heading and the `.card-footer` class adds a footer.

⊛ To add a background color to the card, use contextual classes.

E.g.

```
<div class="card">
  <div class="card-body bg-primary"> Content </div>
</div>
```

⊛ Use `.card-title` to add card title to any heading element. The `.card-text` is used to remove bottom margins for a `<p>` element if it is the last child (or the only one) inside `.card-body`. The `.card-link` class adds a blue color to any link, and a hover effect.

E.g.

```
<div class="card">
  <div class="card-body">
    <h4 class="card-title"> Card title </h4>
    <p class="card-text"> Some text </p>
    <a href="#" class="card-link"> This is link </a>
  </div>
</div>
```

⊛ We can add ~~`.card-img`~~ `.card-img-top` or `.card-img-bottom` to an `<img>` to place the image at the top or at the bottom inside the card.

Note: we added the image outside the `.card-body` to span the entire width.

E.g.

```
<div class="card" style="width: 400px;">
  
  <div class="card-body">
    ...
  </div>
</div>
```


- ① For turn the image as the background of card, use `.card-img-overlay` to add text on top of the image.

E.g.

```
<div class="card" style="width: 400px;">
  
  <div class="card-img-overlay">
    ...
  </div>
</div>
```

Dropdowns

- ① A dropdown menu is a toggleable menu that allows the user to choose one value from a predefined text.

- ② The `.dropdown` indicates a dropdown menu.

* To open the dropdown menu, use a button or a link with a class of `.dropdown-toggle` and the `data-bs-toggle="dropdown"` attribute.

- * Add the `.dropdown-menu` to a container element (like `<div>` or `<ul>`), to actually build the dropdown menu. Then add the `.dropdown-item` to each element (links or buttons) inside the dropdown menu.

E.g.

```
<div class="dropdown">
  <button type="button" class="btn btn-primary dropdown-toggle"
    data-bs-toggle="dropdown"> Dropdown button
  </button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Link 1</a></li>
    ...
  </ul>
</div>
```

- ③ The `.dropdown-divider` is used to separate links inside the dropdown menu with a thin horizontal border.

E.g.

```
<li><hr class="dropdown-divider"></li>
```

- ④ The `.dropdown-header` is used to add headers inside the dropdown menu.

E.g.

```
<li><h5 class="dropdown-header">Dropdown header 1</h5></li>
```

- ⊛ Highlight a specific dropdown item with the `.active` and disable an item in the dropdown menu with the `.disabled`. These classes use as a combination with `.dropdown-item`
- ⊛ We can create a 'dropend' or 'dropstart' menu, by adding `.dropend` or `.dropstart`

E.g.

```
<div class="dropdown dropend">
...
</div>
```

- ⊛ To right align the dropdown menu, add the `.dropdown-menu-end` to element with `.dropdown-menu`

E.g.

```
<div class="dropdown-menu dropdown-menu-end">
```

- ⊛ If we want to dropdown menu expand upwards instead of downwards, change the `<div>` element with `class="dropdown"` to `"dropup"`

- ⊛ The `.dropdown-item-text` is used to add plain text to a dropdown item or used on links to default link styling.

E.g.

```
<ul class="dropdown-menu">
  <li><a class="dropdown-item-text" href="#">Text link </a></li>
  <li><span class="dropdown-item-text">Just text </span></li>
</ul>
```

Collapse

- ⊛ collapse is useful when we want to ~~hide~~ hide and show large amount of content.

- ⊛ The `.collapse` class indicates a collapsible element (for example `<div>`); this is the content that will be shown or hidden with a click of a button.

- ⊛ To control (show/hide) the collapsible content, add the `data-bs-toggle="collapse"` attribute to an `<a>` or a `<button>` element. Then add the `data-bs-target="#id"` attribute to connect the button with the collapsible content. (`<div id="demo">`)

Note: For `<a>` element, we can use the `href` attribute instead of `data-bs-target` attribute

E.g. `<button data-bs-toggle="collapse" data-bs-target="#demo">`
Collapsible
`</button>`

`<div id="demo" class="collapse">`
This is the show/hide text
`</div>`

- ⊛ By default, the collapsible content is hidden; However, we can add `.show` to show the content by default.

E.g. `<div id="demo" class="collapse show">`
This is the show/hide text
`</div>`

● Accordion

- ⊛ We can make a simple accordion with the use of 'collapse' and 'card' components.
- ⊛ Note: Use the `data-bs-parent` attribute to make sure that all collapse elements under the specified parent will be closed when one of the collapsible item is shown.

E.g. `<div id="accordion">`
 `<div class="card">`
 `<div class="card-header">`
 `<a class="btn" data-bs-toggle="collapse" href="#collapseOne">`
 Collapsible Group item 1
 `</a>`
 `</div>`
 `<div id="collapseOne" class="collapse show" data-bs-parent="#accordion">`
 `<div class="card-body">`
 This is the text
 `</div>`
 `</div>`
 `</div>`
`</div>`

Navs

- ① If we want to create a simple horizontal menu, add the `.nav` to a `<ul>` element, followed by `.nav-item` for each `<li>` and add the `.nav-link` to their links.

E.g. `<ul class="nav">`
 `<li class="nav-item">`
 `<a class="nav-link" href="#">Link </a>`
 `</li>`
`</ul>`

- ② Add the `.justify-content-center` and `.justify-content-end` classes to center or right-align the nav.

E.g. `<ul class="nav justify-content-center">`

- ③ Add the `.flex-column` to create a vertical nav.

E.g. `<ul class="nav flex-column">`

- ④ Turn the nav menu into navigation tabs with the `.nav-tabs`. Add the `.active` to the active/current link.

E.g. `<ul class="nav nav-tabs">`
 `<li class="nav-item">`
 `<a class="nav-link active" href="#">Active </a>`
 `</li>`
 `<li class="nav-item">`
 `<a class="nav-link" href="#">Link </a>`
 `</li>`
`</ul>`

- ⑤ To make the tabs toggleable, add the `data-toggle="tab"` attribute to each link. Then add a `.tab-pane` with a unique ID for every tab and wrap them inside a `<div>` element with `.tab-content`

* If you want to tabs fade in-out when clicking, add `.fade` to `.tab-pane`

E.g. `<ul class="nav nav-tabs">`
 `<li class="nav-item">`
 `<a class="nav-link active" data-bs-toggle="tab" href="#Home">`
 Home
 `</a>`
 `</li>`
`</ul>`

⑧ Turn the nav menu into navigation pills with the `.nav-pills`

E.g. `<ul class="nav nav-pills">`

`<li class="nav-item">`

`<a class="nav-link active" href="#">Active </a>`

`</li>`

`</ul>`

⑨ To make the pills toggleable, add the `data-toggle="pill"` attribute.

E.g. `<ul class="nav nav-pills">`

`<li class="nav-item">`

`<a class="nav-link active" data-bs-toggle="pill" href="#home">`

Home

`</a>`

`</li>`

`<li class="nav-item">`

`<a class="nav-link" data-bs-toggle="pill" href="#menu1">`

Menu 1

`</a>`

`</li>`

`</ul>`

`<div class="tab-content">`

`<div class="tab-pane container active" id="home">text... </div>`

`<div class="tab-pane container fade" id="home">text... </div>`

`</div>`

⑩ Justify tabs/pills with the `.nav-justified` (equal-width)

E.g. `<ul class="nav nav-pills nav-justified">... </ul>`

Navbar

① A navigation bar is a navigation header that is placed at the top of the page.

② Bootstrap 5, navigation bar can extend or collapse, depending on the screen size.

★ A standard navigation bar is created with the `.navbar`, followed by a responsive collapsing class `.navbar-expand-xxl|xl|lg|md|sm`

★ To add links inside navbar, use either `<ul>` or `<div>` with `class="navbar-nav"`. Then add `<li>` elements with `.nav-item` followed by `<a>` with `.nav-link`

E.g. `<nav class="navbar navbar-expand-sm bg-light">`

```
<div class="container-fluid">
  <ul class="navbar-nav">
    <li class="nav-item">
      <a class="nav-link" href="#">Link 1 </a>
    </li>
    <li class="nav-item">
      ...
    </li>
  </ul>
</div>
</nav>
```

- ⊛ When we remove the `.navbar-expand-*` to create a navigation bar will always be vertical.
- ⊛ Add the `.justify-content-center` to `<nav>` element to center the navbar.
- ⊛ The `.navbar-brand` is used to highlight the brand/logo/project name of the page.
 - ⊛ When using the `.navbar-brand` with images, automatically adjust the image to fit navbar vertically.

E.g. `<nav class="navbar navbar-expand-sm bg-dark navbar-dark">`
`<div class="container-fluid">`
`<a class="navbar-brand" href="#"> Logo </a>`
`</div>`
`</nav>`

- ⊛ We can use `.navbar-text` instead of `.navbar-brand` except links.
- ⊛ The navbar can also be fixed at the top or bottom of the page.
 - ⊛ A fixed navbar stays visible in fixed position independent the page scroll.
 - ⊛ The `.fixed-top` makes the navbar fixed at top.
 - ⊛ The `.fixed-bottom` makes the navbar fixed at bottom.

E.g. `<nav class="navbar navbar-expand-sm bg-dark navbar-dark fixed-top">`
`...`
`</nav>`

- ⊛ Use the `.sticky-top` to make the navbar stay at the top when we scroll and past it.

E.g. `<nav class="navbar navbar-expand-sm bg-dark navbar-dark sticky-top">`
`...`
`</nav>`

Carousel

- Ⓐ The carousel is a slideshow for cycling through elements.

E.g. `<!-- Carousel -->`

```
<div id="demo" class="carousel slide" data-bs-ride="carousel">
```

```
<!-- Indicators/dots -->
```

```
<div class="carousel-indicators">
```

```
<button type="button" data-bs-target="#demo" data-bs-slide-to="0"
  class="active"></button>
```

```
<button type="button" data-bs-target="#demo" data-bs-slide-to="1">
</button>
```

```
</div>
```

```
<!-- The slideshow/carousel -->
```

```
<div class="carousel-inner">
```

```
<div class="carousel-item active">
```

```

```

```
</div>
```

```
<div class="carousel-item">
```

```

```

```
</div>
```

```
</div>
```

```
<!-- Left and Right controls/icons -->
```

```
<button class="carousel-control-prev" type="button"
  data-bs-target="#demo" data-bs-slide="prev">
```

```
<span class="carousel-control-prev-icon"></span>
```

```
</button>
```

```
<button class="carousel-control-next" type="button"
  data-bs-target="#demo" data-bs-slide="next">
```

```
<span class="carousel-control-next-icon"></span>
```

```
</button>
```

```
</div>
```

- Ⓐ Add elements ~~with~~ inside `<div class="carousel-caption">` within each `<div class="carousel-item">` to create a caption for each slide.

Modal

- ⊛ The modal component is a dialog box / popup window that is displayed on the top of the current page.

E.g. <!-- Button to open the modal -->
<button type="button" class="btn btn-primary" data-bs-toggle="modal"
data-bs-target="#myModal">

Open modal

</button>

<!-- The Modal -->

<div class="modal" id="myModal">

<div class="modal-dialog">

<div class="modal-content">

<!-- Modal Header -->

<div class="modal-header">

<h4 class="modal-title">Modal heading </h4>

<button type="button" class="btn-close"
data-bs-dismiss="modal"> </button>

</div>

<!-- Modal body -->

<div class="modal-body">

Modal body

</div>

<!-- Modal footer -->

<div class="modal-footer">

<button type="button" class="btn btn-danger"
data-bs-dismiss="modal"> Close </button>

</div>

</div>

</div>

</div>

- ⊛ Use the .fade to add a fading effect when opening and closing the modal.

E.g. <div class="modal fade"> </div>

- ⊛ We can change the modal size by using .modal-sm, .modal-lg or .modal-xl for small, large or extra large modals.

E.g. `<div class="modal-dialog modal-sm">`

- ★ If we want to span the full screen width and height for modals, we can use `.modal-fullscreen`

E.g. `<div class="modal-dialog modal-fullscreen">`

- ★ We can also use responsive full screen modals with these classes.
 - `.modal-fullscreen-sm-down`, `.modal-fullscreen-sm-down`,
 - `.modal-fullscreen-lg-down`, `.modal-fullscreen-xl-down`, `.modal-fullscreen-xxl-down`

- ★ We can center the modal vertically and horizontally within the page, with `.modal-dialog-centered`

E.g. `<div class="modal-dialog modal-dialog-centered">`

- ★ When we have a lot of content inside the modal, a scrollbar is added to the page.

★ However, it is possible to only scroll inside the modal, instead of the page itself, by adding `.modal-dialog-scrollable` to `.modal-dialog`

E.g. `<div class="modal-dialog modal-dialog-scrollable">`

Tooltips

- ★ Tooltip component is small pop-up box that appears when the user moves the mouse pointer over an element.

- ★ To create a tooltip, add the `data-bs-toggle="tooltip"` attribute to an element.

★ Use the 'title' to specify the text.

E.g. `<button type="button" class="btn btn-primary" data-bs-toggle="tooltip" title="Hooray!"> Hover over me </button>`

Note: Tooltips must be initialized with Javascript to work

★ Use the following Js code;

```
<script>
```

```
var tooltipTriggerList = [].slice.call(document.querySelectorAll  
  ("[data-bs-toggle='tooltip']"));
```

```
var tooltipList = tooltipTriggerList.map(function (tooltipTriggerEl) {  
  return new bootstrap.Tooltip(tooltipTriggerEl);  
});
```

```
</script>
```


④ By default, tooltip will appear on top of the element.

★ We can use data-bs-placement attribute with the value "top", "bottom", "left", "right" to align the tooltip text.

Popover

④ Popover component is similar to the tooltips. It is a pop-up box that appears when the user click on the element. Different is it can contain more content than a tooltip.

④ To create a popover, add the data-bs-toggle = "popover" attribute to an element.

★ Use the "title" attribute to specify the header text of popover and data-bs-content attribute to specify the text that should be displayed inside the popover's body.

E.g. `<button type="button" class="btn btn-primary" data-bs-toggle="popover" title="Popover Header" data-bs-content="Some content inside">`

Toggle Popover

`</button>`

Note: Popovers must be initialized with Javascript to work.

★ Try below Js code;

`<script>`

```
var popoverTriggerList = [].slice.call(document.querySelectorAll  
  ('[data-bs-toggle="popover"]'));
```

```
var popoverList = popoverTriggerList.map(function (popoverTriggerEl) {  
  return new bootstrap.Popover(popoverTriggerEl);  
});
```

`</script>`

④ By default, popover will appear on the right side of the element.

★ To change that use data-bs-placement attribute with "top", "bottom", "left", "right" values.

④ By default, popover is closed when we click on the element again. However, we can use data-bs-trigger = "focus" attribute which will close the popover when clicking outside the element.

- Ⓐ If we want to display the popover when we move the mouse pointer over the element use; `data-bs-trigger="hover"` attribute.

Toasts

- Ⓐ Toast component is like an alert box that is only shown for a couple of seconds when something happens. (when the user clicks on a button, submits a form, ... etc.)
- Ⓐ To create a toast, use the `.toast` class and add a `.toast-header` and a `.toast-body` inside of it.

Note: Toasts are hidden by default. Use the `.show` if you want to display it. To close it, use a `<button>` element and add `data-bs-dismiss="toast"`

E.g.

```
<div class="toast show">
  <div class="toast-header">
    Toast Header
    <button type="button" class="btn-close" data-bs-dismiss="toast">
  </button>
</div>
  <div class="toast-body">
    Some text inside toast body
  </div>
</div>
```

- Ⓐ To show a toast with a click of a button, it must initialize with Javascript. select the specified element and call the `toast()` method.

* Try following Js code;

E.g.

```
<script>
  document.getElementById("toastbtn").onclick = function () {
    var toastElList = [].slice.call(document.querySelectorAll(".toast"));
    var toastList = toastElList.map(function (toastEl) {
      return new bootstrap.Toast(toastEl);
    });
    toastList.forEach(toast => toast.show());
  }
</script>
```

Scrollspy

- ⊗ This used to automatically update links in a navigation list based on "scroll" position.

E.g. `<div data-bs-spy="scroll" data-bs-target=".navbar" data-bs-offset="50">`
`<nav class="navbar navbar-expand-sm bg-dark navbar-dark fixed-top">`
`<div class="container-fluid">`
`<ul class="navbar-nav">`
`<li class="nav-item">`
`<a class="nav-link" href="#section1">Section 1 </a>`
`</li>`
`<li class="nav-item">`
`<a class="nav-link" href="#section2">Section 2 </a>`
`</li>`
`</ul>`
`</div>`
`</nav>`
`<div id="section1" class="container-fluid bg-success" style="padding: 100px 20px 30px 30px">`
`<h1>Section 01 </h1>`
`<p>here the text </p>`
`</div>`
`<div id="section2" class="container-fluid bg-warning" style="padding: 100px 20px 30px 30px">`
`<h1>Section 02 </h1>`
`<p>here another text </p>`
`</div>`
`</div>`

Offcanvas

- ⊗ Offcanvas is similar to 'modals', except that is often used as a sidebar navigation menu.

E.g. `<div class="offcanvas offcanvas-start" id="demo">`
`<div class="offcanvas-header">`
`<h1 class="offcanvas-title">Heading </h1>`
`<button type="button" class="btn-close" data-bs-dismiss="offcanvas">`
`</button>`
`</div>`


```

<div class="offcanvas-body">
  <p> This is text </p>
  <p> This is text </p>
  <button class="btn btn-secondary" type="button"> Button </button>
</div>
</div>

<div class="container-fluid mt-3">
  <h3> Offcanvas Sidebar </h3>
  <p> Click below button to get sidebar. </p>
  <button class="btn btn-primary" type="button" data-bs-toggle="offcanvas"
    data-bs-target="#demo">
    Open sidebar
  </button>
</div>

```

- ⊛ We can use .offcanvas-start/end/top/bottom to position the offcanvas to left, right, top or bottom.

E.g. <div class="offcanvas offcanvas-end" id="demo">

- ⊛ We can also control when we want to hide or show the offcanvas menu on different screen widths, with the .offcanvas-sm/md/lg/xl/xxl

E.g. <div class="offcanvas offcanvas-start offcanvas-lg" id="demo">

- ⊛ Use the .text-bg-dark to create a dark offcanvas menu.

Tip: We also added the .btn-close-white to .btn-close to create white close button that looks nice with the dark background.

E.g. <div class="offcanvas offcanvas-end" id="demo">
 <button type="button" class="btn-close btn-close-white"
 data-bs-dismiss="offcanvas"> </button>

Utilities/helper classes

- ⊛ Use the .border to add or remove borders.

E.g.

`<span class="border border-bottom-0"></span>`

`<span class="border border-start-0"></span>`

`<br>`

`<span class="border-top"></span>`

`<span class="border-end"></span>`

`<span class="border-bottom"></span>`

`<span class="border-start"></span>`

- ★ Use `.border-1` to `.border-5` to change the width of border.

E.g. `<span class="border border-3"></span>`

- ★ Add color to the border with contextual border color classes.

E.g. `<span class="border border-danger"></span>`

- ★ Add rounded corners to an element with the rounded classes.

★ We can use `.rounded`, `.rounded-top`, `.rounded-end`, `.rounded-bottom`,
`.rounded-start`, `.rounded-circle`, `.rounded-pill`, `.rounded-0`,
`.rounded-1`, `.rounded-2`, `.rounded-3`, `.rounded-4`, `.rounded-5`

E.g. `<span class="rounded-top"></span>`

- ★ Float an element to the right with the `.float-end`, to the left with the `.float-start` and clear floatings with `.clearfix`

E.g. `<div class="clearfix">`

`<span class="float-start">Float Left </span>`

`<span class="float-endleft">Float Right </span>`
`</div>`

- ★ We can center an element with the `.mx-auto` class. (adds `margin-left` and `margin-right` → `auto`)

E.g. `<div class="mx-auto bg-warning">Centered </div>`

- ★ Set the width of an element with `.w-25`, `.w-50`, `.w-75`, `.w-100`, `.w-auto`, `.mw-100` classes.

E.g. `<div class="w-50 bg-warning">Width 50% </div>`

- ★ Set the height of an element with `.h-25`, `.h-50`, `.h-75`, `.h-100`, `.h-auto`, `.mh-100` classes.

E.g. `<div class="h-50 bg-warning">Height 50% </div>`

① Use the shadow classes to add shadows to an element.

We can use `.shadow-none`, `.shadow-sm`, `.shadow`, `.shadow-lg`

E.g. `<div class="shadow-sm p-4 mb-4 bg-white">Small shadow </div>`

② Use the `.align-baseline`, `.align-top`, `.align-middle`, `.align-bottom`, `.align-text-top`, `.align-text-bottom` to change the alignment of elements.

E.g. `<span class="align-top">Top </span>`

③ Create responsive video or slideshow based on the width of the parent.
Add the `.ratio` together with an aspect ratio of your choice `.ratio-1x1`, `.ratio-4x3`, `.ratio-16x9`, `.ratio-21x9` and add the embed (video or iframe) inside of it.

E.g. `<div class="ratio ratio-16x9">
 <iframe src="https://..."></iframe>
</div>`

④ Use the `.visible` or `.invisible` classes to control the visibility of an element.

E.g. `<div class="visible bg-warning">I am visible </div>`

Dark Mode

① By default, bootstrap pages have a white (light) bg color.

② If we want to whole page color to darker, then add `data-bs-theme="dark"` to the `<html>` element.

③ If only want to add darker to a specific element, use above attribute to that component.

E.g. `<table class="table" data-bs-theme="dark"> ... </table>`

Flexbox

① To create a flexbox, use the `.d-flex`

E.g. `<div class="d-flex p-3 bg-secondary text-white">
 <div class="p-2 bg-info">Flex item 1 </div>
 <div class="p-2 bg-warning">Flex item 2 </div>
</div>`

- ④ To create an inline flexbox container use, `.d-inline-flex`
- ④ Use `.flex-row` to display the flex items horizontally (side by side)
 * This is default.

And use `.flex-row-reverse` to right align the horizontal direction.

- ④ Use `.flex-column` or `.flex-column-reverse` to add the vertical direction.
- ④ Use the `.justify-content-start`, `.justify-content-end`, `.justify-content-center`, `.justify-content-between`, `.justify-content-around` to change the alignment of the flex items.

E.g. `<div class="d-flex justify-content-between"> ... </div>`

- ④ Use `.flex-fill` on flex items to force them into equal widths.

E.g. `<div class="d-flex">`
 `<div class="p-2 bg-info flex-fill"> ... </div>`
 `</div>`

- ④ Use `.flex-grow-1` on a flex item to take up the rest of the space.

E.g. `<div class="d-flex">`
 `<div class="p-2 bg-info"> Flex item 1 </div>`
 `<div class="p-2 bg-info flex-grow-1"> Flex item 2 </div>`
 `</div>`